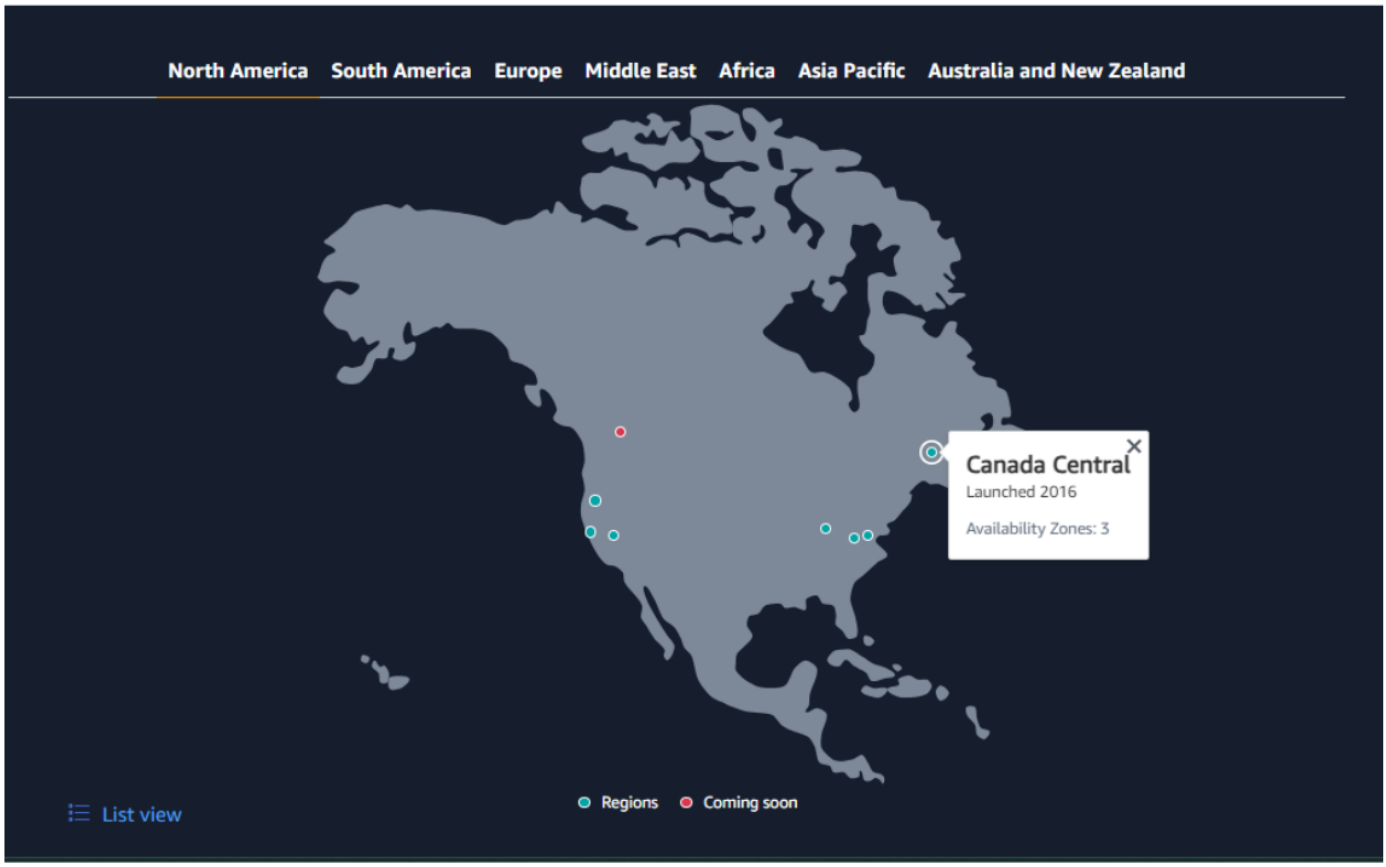
Amazon Cloud

Global Infastructure: https://aws.amazon.com/about-aws/global-infrastructure/regions_az/ (https://aws.amazon.com/about-aws/global-infrastructure/regions_az/)



Regions

AWS has the concept of a Region, which is a physical location around the world where we cluster data centers. We call each group of logical data centers an Availability Zone. Each AWS Region consists of a minimum of three, isolated, and physically separate AZs within a geographic area. Unlike other cloud providers, who often define a region as a single data center, the multiple AZ design of every AWS Region offers advantages for customers. Each AZ has independent power, cooling, and physical security and is connected via redundant, ultra-low-latency networks. AWS customers focused on high availability can design their applications to run in multiple AZs to achieve even greater fault-tolerance. AWS infrastructure Regions meet the highest levels of security, compliance, and data protection.

AWS provides a more extensive global footprint than any other cloud provider, and to support its global footprint and ensure customers are served across the world, AWS opens new Regions rapidly. AWS maintains multiple geographic Regions, including Regions in North America, South America, Europe, China, Asia Pacific, South Africa, and the Middle East.

Show less

Availability Zones

An Availability Zone (AZ) is one or more discrete data centers with redundant power, networking, and connectivity in an AWS Region. AZs give customers the ability to operate production applications and databases that are more highly available, fault tolerant, and scalable than would be possible from a single data center. All AZs in an AWS Region are interconnected with high-bandwidth, low-latency networking, over fully redundant, dedicated metro fiber providing high-throughput, low-latency networking between AZs. All traffic between AZs is encrypted. The network performance is sufficient to accomplish synchronous replication between AZs. AZs make partitioning applications for high availability easy. If an application is partitioned across AZs, companies are better isolated and protected from issues such as power outages, lightning strikes, tornadoes, earthquakes, and more. AZs are physically separated by a meaningful distance, many kilometers, from any other AZ, although all are within 100 km (60 miles) of each other.

Show less

Services

AWS offers a broad set of global cloud-based products including compute, storage, database, analytics, networking, machine learning and AI, mobile, developer tools, IoT, security, enterprise applications, and much more.

The following core services are included in all Region launches: Amazon API Gateway, AWS Application Auto Scaling, Amazon Aurora, AWS Certificate Manager (ACM), AWS CloudFormation, AWS CloudTrail, Amazon CloudWatch, AWS CodeDeploy, AWS Config, AWS Database Migration Service (AWS DMS), AWS Direct Connect, Amazon DynamoDB, Amazon EC2 Auto Scaling, Amazon Elastic Block Store (Amazon EBS), Amazon Elastic Compute Cloud (Amazon EC2), Amazon Elastic Container Registry (Amazon ECR), Amazon Elastic Container Service (Amazon ECS), Elastic Load Balancing (ELB), Amazon EMR, Amazon ElastiCache, Amazon EventBridge, AWS Identity and Access Management (IAM), AWS Key Management Service (AWS KMS), Amazon Kinesis, Data Streams, AWS Lambda, AWS Management Console, AWS Marketplace, Amazon OpenSearch Service, AWS Health Dashboard, Amazon Redshift, Amazon Relational Database Service (Amazon RDS), Amazon Route 53, AWS Security Token Service (AWS STS), Amazon Simple Notification Service (Amazon SNS), Amazon Simple Queue Service (Amazon SQS), Amazon Simple Storage Service (Amazon S3), Amazon Simple Workflow Service (Amazon SWF), AWS Step Functions, AWS Support, AWS Systems Manager, AWS Trusted Advisor, Amazon Virtual Private Cloud (Amazon VPC), and AWS VPN.

image.png

By Use Case
By Industry
By Organization Type

By Use Case

Archiving
Affordable solutions for data archiving from gigabytes to petabytes

Backup and Restore
Durable, cost-effective options for backup and disaster recovery

Blockchain
Shared ledgers for trusted transactions among multiple parties

Cloud Migration
Easily migrate apps and data to AWS

Cloud Operations
Operate securely and safely in the cloud, at scale

Containers
Fully managed services for every workload

Content Delivery
Accelerate websites, APIs, and video content

Database Migrations
Save time and cost by migrating to fully managed databases

Data Lakes and Analytics

Compute (https://aws.amazon.com/free/?all-free-tier.sort-by=item.additionalFields.SortRank&all-free-tier.sort-order=asc&awsf.Free%20Tier%20Types=*all&awsf.Free%20Tier%20Categories=categories%23compute)

Database (https://aws.amazon.com/free/?all-free-tier.sort-by=item.additionalFields.SortRank&all-free-tier.sort-order=asc&awsf.Free%20Tier%20Types=*all&awsf.Free%20Tier%20Categories=categories%23databases&awsf.tech-category=tech-category%23storage)

Storage (https://aws.amazon.com/free/?all-free-tier.sort-by=item.additionalFields.SortRank&all-free-tier.sort-order=asc&awsf.Free%20Tier%20Types=*all&awsf.Free%20Tier%20Categories=categories%23storage&awsf.tech-category=tech-category%23storage)

Containers (https://aws.amazon.com/free/?all-free-tier.sort-by=item.additionalFields.SortRank&all-free-tier.sort-

order=asc&awsf.Free%20Tier%20Types=*all&awsf.Free%20Tier%20Categories=categories%23containers&awsf.tech-category=tech-category%23storage)

Web and Mobile App (https://aws.amazon.com/free/?all-free-tier.sort-by=item.additionalFields.SortRank&all-free-tier.sort-order=asc&awsf.Free%20Tier%20Types=*all&awsf.Free%20Tier%20Categories=categories%23mobile&awsf.tech-category=tech-category%23storage)

Serverless (https://aws.amazon.com/free/?all-free-tier.sort-by=item.additionalFields.SortRank&all-free-tier.sort-order=asc&awsf.Free%20Tier%20Types=*all&awsf.Free%20Tier%20Categories=categories%23serverless&awsf.tech-category=tech-category%23storage)

Machine Learning (https://aws.amazon.com/free/?all-free-tier.sort-by=item.additionalFields.SortRank&all-free-tier.sort-order=asc&awsf.Free%20Tier%20Types=*all&awsf.Free%20Tier%20Categories=categories%23ai-ml)